E-NGBoost: Evidential Natural Gradient Boosting for Tabular Uncertainty Quantification

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Abstract

Reliable uncertainty estimates for tabular regression are important for decision-making under distribution shift, but high-performing baselines such as deep ensembles can be computationally expensive. We study a simple, single-model alternative that combines Natural Gradient Boosting (NGBoost) with a Normal–Inverse–Gamma (NIG) evidential parameterization, which we call E-NGBoost. The resulting predictive distribution is a Student-t marginal with a closed-form decomposition into aleatoric and epistemic components.

To improve robustness and encourage lower evidence in low-density regions, we train with a penalized Student-t negative log-likelihood augmented with (i) evidential regularization terms and (ii) a density-aware evidence prior based on Isolation Forest fit on the training split only. Since such modifications are not strictly proper scoring rules in general, we emphasize evaluation using strictly proper metrics (CRPS and held-out NLL) and calibration diagnostics, and we compare against distribution-free interval baselines (Split Conformal, CQR, QRF).

Across several OpenML tabular datasets, E-NGBoost is competitive with NGBoost in-distribution and provides useful uncertainty ranking signals for feature-aligned OOD detection in some settings. We also observe practical limitations including hyperparameter sensitivity and shift dependence, and position evidential "epistemic" uncertainty as a relative score useful for OOD detection and selective prediction rather than an absolutely calibrated posterior uncertainty estimate.

1 Introduction

Tabular regression appears in many high-stakes applications (pricing, risk, healthcare), where point accuracy alone is insufficient: we want uncertainty that is *calibrated* and helpful for *risk-aware* decisions. On tabular data, tree/boosting methods often remain strong due to their ability to capture complex feature interactions, yet uncertainty quantification (UQ) for boosted trees is commonly approached via ensembles or post-hoc calibration. While effective, these approaches can be computationally expensive (ensembles require training multiple models) or require additional assumptions about the calibration process.

We study a simple single-model UQ approach within the NGBoost framework (5) by replacing the Gaussian likelihood with an evidential Normal–Inverse–Gamma (NIG) parameterization (1). This yields a Student-t predictive marginal with an analytic decomposition into (i) aleatoric uncertainty (data noise) and (ii) epistemic uncertainty (evidence/virtual observations). The epistemic component is not a guaranteed calibrated posterior uncertainty; we therefore treat it as a *relative* signal and evaluate its usefulness through proper metrics and downstream behavior rather than strong theoretical claims.

To strengthen robustness and to encourage lower evidence in low-density regions, we incorporate a density proxy via Isolation Forest (8) fit on the training split only. We use this proxy both as (i) a training-time evidence regularizer and (ii) an optional inference-time evidence calibration. Since these modifications introduce prediction-dependent weights and regularizers, the resulting training objective is not a strictly proper scoring rule in general. We therefore emphasize evaluation using proper held-out metrics (CRPS and held-out NLL) and calibration diagnostics (6).

Contributions.

1. **E-NGBoost:** a practical integration of NGBoost with NIG evidential regression, yielding Student-t predictive marginals and an uncertainty decomposition for tabular regression.
2. **Density-aware evidential boosting:** a simple density-aware inductive bias that encourages lower evidence in low-density regions, using Isolation Forest fit on train only.
3. **Leakage-controlled evaluation:** a conservative experimental protocol that reports proper scoring rules (CRPS and held-out NLL) alongside calibration and risk metrics, and compares against distribution-free interval baselines (Split Conformal, CQR, QRF).

2 Background

2.1 NGBoost

NGBoost (5) extends gradient boosting to probabilistic prediction by learning the parameters of a chosen predictive distribution. At each boosting iteration, base learners are fit to the (preconditioned) gradient of a scoring rule with respect to distribution parameters. The key idea is to use a natural-gradient update, which preconditions the Euclidean gradient by the Fisher information matrix of the predictive distribution, improving optimization behavior across parameterizations.

2.2 Normal–Inverse–Gamma and Student-t predictive marginal

Evidential regression with a Normal–Inverse–Gamma (NIG) parameterization (1) models predictive uncertainty by placing a conjugate prior over the Gaussian mean and variance. With NIG parameters $(\gamma, \nu, \alpha, \beta)$, marginalizing the latent mean and variance yields a Student-t predictive distribution. A commonly used variance decomposition is

$$\mathrm{Var}_{al} = \mathbb{E}[\sigma^2] = \frac{\beta}{\alpha - 1}, \qquad \mathrm{Var}_{ep} = \mathrm{Var}[\mu] = \frac{\beta}{\nu(\alpha - 1)}.$$

We use these as diagnostic components. Following recent analyses that critique evidential epistemic uncertainty, we treat "epistemic" primarily as a relative score and validate it empirically rather than claiming that it is an absolutely calibrated posterior quantity (4; 7; 13).

2.3 Proper scoring rules and calibration diagnostics

Proper scoring rules provide principled incentives for probabilistic forecasts. In this work we report the negative log-likelihood (log score) and the continuous ranked probability score (CRPS), which is strictly proper for univariate distributions (6). We complement proper scores with calibration diagnostics based on prediction intervals (coverage and width) and PIT-based diagnostics, which can reveal systematic miscalibration patterns even when point accuracy is similar.

3 Method: E-NGBoost

3.1 Predictive distribution and parameterization

For each input x, E-NGBoost predicts NIG parameters $(\gamma, \nu, \alpha, \beta)$. The predictive marginal is Student-t with degrees of freedom 2α, location γ, and scale

$$s = \sqrt{\frac{\beta(1 + \nu)}{\nu\alpha}}.$$

For numerical stability and to respect constraints ($\nu > 0$, $\alpha > 1$, $\beta > 0$), we optimize unconstrained variables by using a log-parameterization for constrained parameters:

$$(\gamma, \log \nu, \log(\alpha - 1), \log \beta).$$

3.2 Training objective: penalized Student-t NLL

Our training objective combines the Student-t negative log-likelihood with several regularization terms designed to improve robustness and encourage desirable uncertainty behavior:

$$\mathcal{L} = (1 - w_{\text{mae}})\, w_{\text{nll}}\, \mathcal{L}^{\text{NLL}} + w_{\text{mae}}\, |y - \gamma| + \lambda_r\, \mathcal{L}^{R}_{\text{norm}} + \lambda_u\, \mathcal{L}^{U} + \lambda_d\, \mathcal{L}^{\text{density}}. \tag{1}$$

Core likelihood term. $\mathcal{L}^{\text{NLL}}$ is the closed-form Student-t marginal NLL induced by NIG (1).

Variance-weighted likelihood and MAE fallback. We use a stop-gradient variance weight

$$w_{\text{nll}} = \text{sg}[\text{Var}(Y \mid x)]^{\kappa},$$

and treat it as an optimization heuristic rather than a scoring rule. We also include an MAE term as a robust fallback when the likelihood term becomes unstable.

Aleatoric-normalized evidential regularizer. We use an aleatoric-normalized evidential regularizer

$$\mathcal{L}^{R}_{\text{norm}} = \frac{|y - \gamma|}{\sqrt{\beta/(\alpha - 1)}} \cdot (2\nu + \alpha),$$

inspired by normalization ideas in prior work (10). Unlike normalization based on the Student-t scale s, this variant uses the expected aleatoric standard deviation $\sqrt{\beta/(\alpha - 1)}$ and does not involve ν in the normalization.

Auxiliary stability term. We include

$$\mathcal{L}^{U} = \frac{(y - \gamma)^2}{\alpha},$$

whose gradient w.r.t. α decays as $O(1/\alpha^2)$; we do not claim equivalence to other uncertainty-aware objectives and select λ_u using validation CRPS.

Properness boundary. The Student-t log score alone is strictly proper, but adding prediction-dependent weights and regularizers generally breaks strict properness of the training loss. We therefore treat (1) as an optimization surrogate and rely on proper held-out metrics (CRPS and held-out NLL) and calibration diagnostics for model selection and reporting (6).

3.3 Density-aware evidence regularization and inference-time calibration

We fit an Isolation Forest density proxy $\text{dens}(x) \in [0, 1]$ *only on the training split* (8). During training, we regularize evidence as

$$\nu_{\text{target}}(x) = s_\nu \cdot \text{dens}(x), \qquad \mathcal{L}^{\text{density}} = \frac{(\nu - \nu_{\text{target}})^2}{1 + \nu_{\text{target}}}.$$

This is a heuristic inductive bias inspired by distance/density awareness for UQ (9): we encourage lower evidence in low-density regions, but do not claim formal guarantees.

Optionally at inference time, we calibrate evidence by scaling ν with the query density:

$$\nu_{\text{cal}}(x) = \nu(x) \cdot \frac{\text{dens}(x)}{\mathbb{E}_{train}[\text{dens}]},$$

and compute epistemic variance using ν_{cal} to strengthen OOD ranking signals in low-density regions.

3.4 Natural gradient implementation (diagonal Fisher)

E-NGBoost uses the NGBoost natural gradient update. We implement analytic diagonal Fisher terms for the NIG parameterization and perform updates in log-parameter space. For completeness, we provide the core mathematical formulation and the diagonal Fisher expressions in Appendix A.

4 EXPERIMENTAL SETUP

4.1 DATASETS

We evaluate on standard tabular regression datasets: CONCRETE COMPRESSIVE STRENGTH (OpenML), ENERGY EFFICIENCY (OpenML), BOSTON HOUSING (OpenML), KIN8NM (OpenML), and DIABETES (sklearn).

4.2 BASELINES

We compare against:

1. **NGBoost** with a Gaussian likelihood (5).
2. **Ensemble(5× NGBoost)** (a mixture evaluation for NLL/CRPS).
3. **CatBoost-UQ** (CatBoost regression with built-in uncertainty output).
4. **Interval baselines:** Split Conformal (2), Conformalized Quantile Regression (CQR) (12), and Quantile Regression Forests (QRF) (11).

4.3 EVALUATION PROTOCOL AND METRICS

We use a conservative, leakage-controlled protocol:

1. We run 5 random splits (seeded) with test size 0.2.
2. All preprocessing (e.g., feature scaling) is fit on the training split only.
3. The Isolation Forest density proxy (when used) is fit on the training split only (8).

We report in-distribution (ID) metrics: RMSE, NLL, CRPS, coverage (PICP@90), width (MPIW@90), and PIT-ECE. Interval-only methods do not define likelihood/PIT-based metrics; these entries are reported as "–".

Regularization selection. When using evidential regularization hyperparameters $(\lambda_r, \lambda_u, \lambda_d)$, a practical selection rule is to use a held-out calibration split and select by calibration CRPS (train/cal/test $= 60/20/20$), then retrain on train+cal and evaluate on test. We demonstrate this protocol explicitly on ENERGY.

4.4 E-NGBOOST VARIANTS (ABLATION STUDY)

To clarify the roles of different loss terms, we report several E-NGBoost configurations:

1. **(A) Base:** Student-t NLL + MAE fallback, without additional evidence/density regularizers.
2. **(B) Evidential regularization only:** enables λ_r, λ_u while disabling density regularization ($\lambda_d = 0$).
3. **(C) Density regularization only:** enables λ_d while disabling evidential regularization ($\lambda_r = \lambda_u = 0$).
4. **(D) Full:** enables both evidential and density regularization terms.

We also include a **robust-only** configuration that removes the variance-weighted likelihood term but keeps the MAE fallback, and an **Energy tuned** configuration selected by calibration-CRPS.

4.5 OOD SHIFTS AND DETECTION METRICS

For OOD evaluation we use feature-aligned shifts that preserve the feature space:

- **tail:** hold out the upper tail of one feature;
- **domain:** hold out rare categories of a discrete-like feature (when available);

- **perturb:** add feature noise (no labels needed).

OOD detection uses uncertainty scores (variance proxies) and reports AUROC, FPR@95, and AUPR; we also report selective prediction metrics (AURC and Risk@80% coverage) on the ID test set.

5 RESULTS

5.1 IN-DISTRIBUTION PROBABILISTIC REGRESSION

Table 1 reports in-distribution metrics (RMSE/NLL/CRPS plus coverage–width and PIT-ECE). Interval-only methods do not define likelihood- or PIT-based metrics, so these entries are shown as "–". To improve readability, we bold the best values among probabilistic methods (excluding interval-only baselines) for RMSE/NLL/CRPS/PIT-ECE.

Some regularization-only configurations can cause severe over-coverage (PICP$\approx$ 1) or unstable NLL (e.g., density-only variants on Boston/Kin8nm), illustrating hyperparameter sensitivity.

Student-t heavy tails and robustness. Student-t marginals can improve distributional fit even when point accuracy is similar. For example on CONCRETE, E-NGBoost (A) matches NGBoost in RMSE (4.76 vs. 4.78) while improving NLL (3.19 vs. 3.26), consistent with heavier tails better accommodating outliers without sacrificing mean prediction quality.

5.2 OOD DETECTION (FEATURE-ALIGNED SHIFTS) AND SELECTIVE PREDICTION

Table 2 summarizes OOD detection and selective prediction metrics under feature-aligned shifts. We highlight three observations:

1. **Boston tail shift.** E-NGBoost shows strong OOD discrimination (AUROC 0.942, FPR@95 0.246), outperforming both NGBoost and the 5$\times$ NGBoost ensemble under this protocol.
2. **Energy tail shift.** All methods are near chance (AUROC around 0.25–0.43), suggesting this shift is not captured well by the uncertainty proxies used here (a limitation).
3. **Kin8nm perturb shift.** The ensemble remains the strongest detector, while E-NGBoost is competitive with NGBoost.

5.3 BOUNDARY OF APPLICABILITY AND PRACTICAL CONSIDERATIONS

1. **Hyperparameter sensitivity (and a default mitigation).** On Energy, some evidential regularizers strongly over-cover (PICP$\approx$ 1) and inflate CRPS. A practical default is to select regularization strengths on a held-out calibration split by minimizing calibration-CRPS.
2. **Density-only instability.** In some settings, density regularization alone can lead to unstable NLL (large variance across splits), suggesting non-trivial interactions with other loss terms.
3. **Shift dependence.** Epistemic proxies can be informative for some feature-aligned shifts (e.g., Boston tail; Kin8nm perturb), but fail on others (e.g., Energy tail). We therefore position evidential "epistemic" as a relative score rather than a universal OOD detector.

6 LIMITATIONS AND FUTURE WORK

Limitations.

1. **Training loss is not strictly proper.** Once prediction-dependent weights and regularizers are added, the training objective is no longer a strictly proper scoring rule in general. We therefore do not claim incentive guarantees from the training loss and rely on proper held-out metrics (CRPS and held-out NLL) plus calibration diagnostics for reporting.

Table 1: In-distribution regression results (mean±std over 5 splits). Best values among probabilistic methods (excluding interval-only baselines) are bolded for RMSE/NLL/CRPS/PIT-ECE.

Dataset	Method	RMSE ↓	NLL ↓	CRPS ↓	PICP@90	MPIW@90	PIT-ECE ↓
	NGBoost	4.7771 ± 0.5833	3.2553 ± 0.2371	2.4993 ± 0.2538	0.734 ± 0.029	8.8775 ± 0.2629	0.3814 ± 0.0401
	Ensemble(5x NGBoost)	**4.6905 ± 0.5960**	**3.0925 ± 0.2103**	**2.4539 ± 0.2600**	0.768 ± 0.023	9.2784 ± 0.1933	**0.3476 ± 0.0373**
	CatBoost-UQ	4.7617 ± 0.3897	4.2155 ± 0.6227	2.5086 ± 0.1755	0.697 ± 0.020	7.7636 ± 0.1882	0.5204 ± 0.0205
CONCRETE	E-NGBoost (A)	4.7629 ± 0.4929	3.1856 ± 0.2242	2.4994 ± 0.2203	0.733 ± 0.025	9.0561 ± 0.2172	0.3612 ± 0.0701
	SplitConformal (CatBoost, cal=0.2)	5.0810 ± 0.4847	–	–	0.906 ± 0.018	15.9028 ± 0.7726	–
	CQR (CatBoost, cal=0.2)	4.5375 ± 0.6016	–	–	0.901 ± 0.019	19.2579 ± 0.7399	–
	QRF (quantile-forest, cal=0.2)	5.7841 ± 0.3968	–	3.0679 ± 0.1615	0.926 ± 0.014	24.5510 ± 0.5040	–
	NGBoost	0.3770 ± 0.0294	0.3392 ± 0.0707	0.2037 ± 0.0155	0.788 ± 0.013	0.9318 ± 0.0347	0.3714 ± 0.0729
	Ensemble(5x NGBoost)	0.3760 ± 0.0333	**0.2723 ± 0.0889**	**0.2009 ± 0.0176**	0.812 ± 0.024	0.9551 ± 0.0262	**0.3257 ± 0.0760**
	CatBoost-UQ	0.4251 ± 0.0485	1.4286 ± 0.5572	0.2376 ± 0.0300	0.583 ± 0.036	0.6916 ± 0.0512	0.6758 ± 0.0821
	E-NGBoost (A)	0.3788 ± 0.0288	0.3243 ± 0.0760	0.2034 ± 0.0150	0.805 ± 0.012	0.9365 ± 0.0312	0.3377 ± 0.0600
	E-NGBoost (B-evidreg)	0.3869 ± 0.0309	1.0511 ± 0.0171	0.3096 ± 0.0081	1.000 ± 0.000	3.6119 ± 0.0614	0.9958 ± 0.0489
	E-NGBoost (C-density)	0.3789 ± 0.0350	0.3344 ± 0.0963	0.2035 ± 0.0185	0.791 ± 0.029	0.9366 ± 0.0310	0.3351 ± 0.0560
ENERGY	E-NGBoost (D-full)	**0.3745 ± 0.0420**	1.4931 ± 0.0088	0.4472 ± 0.0071	1.000 ± 0.000	5.7845 ± 0.0344	1.2525 ± 0.0534
	E-NGBoost (robust-only)	0.4595 ± 0.0556	1.3049 ± 0.0170	0.3883 ± 0.0146	0.999 ± 0.003	4.5499 ± 0.0532	0.9860 ± 0.0576
	SplitConformal (CatBoost, cal=0.2)	0.5045 ± 0.0558	–	–	0.918 ± 0.028	1.8272 ± 0.1456	–
	CQR (CatBoost, cal=0.2)	0.3891 ± 0.0547	–	–	0.903 ± 0.040	2.4208 ± 0.4889	–
	QRF (quantile-forest, cal=0.2)	0.6935 ± 0.0580	–	0.3360 ± 0.0217	0.883 ± 0.009	3.9355 ± 0.1697	–
	E-NGBoost (energy tuned)	0.3791 ± 0.0283	0.3379 ± 0.0731	0.2039 ± 0.0143	0.805 ± 0.027	0.9404 ± 0.0344	0.3491 ± 0.0356
	NGBoost	2.7506 ± 0.1778	6.5004 ± 1.0702	1.6767 ± 0.0978	0.451 ± 0.021	2.8591 ± 0.0598	0.8745 ± 0.0124
	Ensemble(5x NGBoost)	**2.6467 ± 0.1198**	4.2346 ± 0.3670	1.5330 ± 0.0692	0.563 ± 0.016	3.4895 ± 0.1195	0.6761 ± 0.0229
	CatBoost-UQ	2.7945 ± 0.2123	10.7265 ± 1.4856	1.6569 ± 0.1315	0.459 ± 0.041	2.8060 ± 0.1715	0.8824 ± 0.0828
	E-NGBoost (A)	2.8612 ± 0.1482	6.8297 ± 0.9698	1.6661 ± 0.0201	0.504 ± 0.023	3.1358 ± 0.1281	0.8031 ± 0.0739
	E-NGBoost (B-evidreg)	2.7123 ± 0.1210	2.6452 ± 0.1973	**1.4781 ± 0.0489**	0.775 ± 0.032	5.8672 ± 0.1973	0.3208 ± 0.0647
BOSTON	E-NGBoost (C-density)	2.8914 ± 0.1419	123.3550 ± 235.1404	1.6579 ± 0.0505	0.553 ± 0.035	3.3807 ± 0.3014	0.6847 ± 0.0647
	E-NGBoost (D-full)	2.8160 ± 0.1673	**2.4305 ± 0.0738**	1.4921 ± 0.0899	0.876 ± 0.031	8.3440 ± 0.1812	**0.2227 ± 0.0564**
	E-NGBoost (robust-only)	2.9673 ± 0.2584	3.8088 ± 0.4637	1.6577 ± 0.1209	0.604 ± 0.034	4.0333 ± 0.3005	0.6259 ± 0.0721
	SplitConformal (CatBoost, cal=0.2)	3.0631 ± 0.2351	–	–	0.937 ± 0.033	12.6104 ± 2.7570	–
	CQR (CatBoost, cal=0.2)	2.9995 ± 0.1167	–	–	0.941 ± 0.028	14.2001 ± 1.7503	–
	QRF (quantile-forest, cal=0.2)	3.2411 ± 0.2745	–	1.6804 ± 0.1049	0.935 ± 0.034	12.5816 ± 1.1002	–
	NGBoost	0.1658 ± 0.0023	-0.4661 ± 0.0212	0.0912 ± 0.0015	0.834 ± 0.009	0.4371 ± 0.0033	0.2677 ± 0.0176
	Ensemble(5x NGBoost)	0.1646 ± 0.0023	-0.4828 ± 0.0176	0.0902 ± 0.0015	0.846 ± 0.008	0.4401 ± 0.0021	0.2550 ± 0.0164
	CatBoost-UQ	**0.1149 ± 0.0032**	**-0.7153 ± 0.0481**	**0.0638 ± 0.0017**	0.781 ± 0.010	0.2719 ± 0.0049	0.3301 ± 0.0291
	E-NGBoost (A)	0.1647 ± 0.0026	-0.4588 ± 0.0229	0.0903 ± 0.0017	0.824 ± 0.009	0.4306 ± 0.0050	0.2843 ± 0.0163
	E-NGBoost (B-evidreg)	0.1653 ± 0.0024	-0.4822 ± 0.0170	0.0899 ± 0.0016	0.893 ± 0.007	0.4911 ± 0.0030	**0.1842 ± 0.0191**
KIN8NM	E-NGBoost (C-density)	0.1690 ± 0.0045	246.6091 ± 366.6946	0.0927 ± 0.0027	0.845 ± 0.019	0.4626 ± 0.0299	0.2685 ± 0.0357
	E-NGBoost (D-full)	0.1726 ± 0.0025	-0.3905 ± 0.0163	0.0951 ± 0.0015	0.946 ± 0.001	0.6140 ± 0.0143	0.1850 ± 0.0167
	E-NGBoost (robust-only)	0.1684 ± 0.0034	-0.4241 ± 0.0244	0.0931 ± 0.0021	0.855 ± 0.008	0.4675 ± 0.0080	0.2385 ± 0.0187
	SplitConformal (CatBoost, cal=0.2)	0.1183 ± 0.0028	–	–	0.896 ± 0.013	0.3789 ± 0.0076	–
	CQR (CatBoost, cal=0.2)	0.1096 ± 0.0033	–	–	0.899 ± 0.012	0.4785 ± 0.0150	–
	QRF (quantile-forest, cal=0.2)	0.1512 ± 0.0022	–	0.0830 ± 0.0012	0.955 ± 0.004	0.6322 ± 0.0023	–
	NGBoost	57.4353 ± 1.9946	8.7623 ± 0.5672	37.2832 ± 1.6028	0.396 ± 0.030	64.2546 ± 0.9677	0.9708 ± 0.0667
	Ensemble(5x NGBoost)	57.4525 ± 1.6328	**7.4962 ± 0.3665**	**36.5416 ± 1.2079**	0.454 ± 0.047	71.3728 ± 1.2264	**0.8908 ± 0.0886**
	CatBoost-UQ	**55.0597 ± 2.5741**	17.3736 ± 8.4968	36.9923 ± 2.6379	0.299 ± 0.065	50.7393 ± 4.1489	1.1281 ± 0.0841
DIABETES	E-NGBoost (A)	57.3690 ± 2.4365	8.3753 ± 0.3227	37.2491 ± 1.6118	0.438 ± 0.034	66.5322 ± 1.6012	0.9717 ± 0.0518
	SplitConformal (CatBoost, cal=0.2)	55.1766 ± 2.6142	–	–	0.942 ± 0.046	203.8124 ± 18.2553	–
	CQR (CatBoost, cal=0.2)	55.8810 ± 3.2073	–	–	0.912 ± 0.050	200.1184 ± 19.0924	–
	QRF (quantile-forest, cal=0.2)	55.1662 ± 3.3714	–	31.2877 ± 1.7119	0.915 ± 0.039	178.2687 ± 6.9824	–

2. **Evidential "epistemic" uncertainty is relative.** Prior analyses show that evidential epistemic uncertainty can behave like an OOD detector rather than a calibrated posterior uncertainty estimate; we position it as a relative score and validate its usefulness through OOD detection and selective prediction behavior (4; 7; 13).

3. **Hyperparameter sensitivity and instability.** Some regularization settings can lead to severe over-coverage (PICP≈ 1) or unstable NLL (large variance across splits), especially for density-only configurations. We partially mitigate this by recommending calibration-CRPS selection when regularization is enabled.

4. **Shift dependence.** Density-aware evidence is expected to react when a shift moves inputs away from the training support. It may be less informative for shifts that keep x within the training support while changing $p(y \mid x)$; our Energy tail results reflect this limitation.

Table 2: OOD detection and selective prediction under feature-aligned shifts (mean±std over 5 splits). We report AUROC/FPR@95/AUPR for OOD detection and AURC/Risk@80 for selective prediction on the ID test set.

Dataset	Shift	Method	AUROC ↑	FPR@95 ↓	AUPR ↑	AURC ↓	Risk@80 ↓	ID	OOD
BOSTON	tail	NGBoost	0.692 ± 0.246	0.587 ± 0.272	0.590 ± 0.269	5.1999 ± 1.2708	6.0672 ± 1.5552	69	33
		Ensemble(5x NGBoost)	0.777 ± 0.080	0.625 ± 0.241	0.597 ± 0.098	**3.6384 ± 0.6926**	**4.7682 ± 1.1678**	69	33
		E-NGBoost (NIG)	**0.942 ± 0.043**	**0.246 ± 0.155**	**0.906 ± 0.073**	4.6243 ± 1.5199	5.4340 ± 1.6727	69	33
CONCRETE	tail	NGBoost	0.755 ± 0.141	0.649 ± 0.166	0.504 ± 0.171	**9.8394 ± 1.9393**	15.4156 ± 4.1035	197	56
		Ensemble(5x NGBoost)	**0.940 ± 0.022**	**0.400 ± 0.251**	**0.849 ± 0.025**	12.0928 ± 1.8062	**13.4103 ± 1.0576**	197	56
		CatBoost-UQ	0.520 ± 0.098	0.878 ± 0.106	0.243 ± 0.055	11.3909 ± 3.7876	13.5862 ± 2.4748	197	56
		E-NGBoost (NIG)	0.798 ± 0.153	0.643 ± 0.232	0.646 ± 0.151	11.5698 ± 1.8838	15.6895 ± 2.9689	197	56
ENERGY	domain	NGBoost	0.703 ± 0.033	0.561 ± 0.138	0.351 ± 0.051	**0.0459 ± 0.0084**	**0.0764 ± 0.0197**	116	38
		Ensemble(5x NGBoost)	**0.831 ± 0.026**	0.820 ± 0.052	**0.801 ± 0.036**	0.0832 ± 0.0190	0.1100 ± 0.0248	116	38
		CatBoost-UQ	0.675 ± 0.026	0.860 ± 0.096	0.380 ± 0.044	0.0589 ± 0.0176	0.0916 ± 0.0242	116	38
		E-NGBoost (NIG)	0.701 ± 0.047	**0.483 ± 0.038**	0.363 ± 0.057	0.0464 ± 0.0113	0.0815 ± 0.0237	116	38
ENERGY	tail	NGBoost	0.268 ± 0.016	0.918 ± 0.028	0.164 ± 0.056	**0.0624 ± 0.0056**	0.1156 ± 0.0141	143	64
		Ensemble(5x NGBoost)	**0.429 ± 0.051**	0.945 ± 0.042	**0.207 ± 0.081**	0.0985 ± 0.0133	0.1336 ± 0.0230	143	64
		CatBoost-UQ	0.249 ± 0.034	0.959 ± 0.025	0.162 ± 0.061	0.0730 ± 0.0141	0.1275 ± 0.0341	143	64
		E-NGBoost (NIG)	0.346 ± 0.048	**0.870 ± 0.057**	0.194 ± 0.036	0.0630 ± 0.0053	**0.1143 ± 0.0168**	143	64
KIN8NM	perturb	NGBoost	0.838 ± 0.033	0.708 ± 0.067	0.861 ± 0.031	0.0136 ± 0.0005	0.0187 ± 0.0006	1639	1639
		Ensemble(5x NGBoost)	**0.925 ± 0.007**	**0.453 ± 0.052**	**0.938 ± 0.006**	0.0208 ± 0.0004	0.0245 ± 0.0009	1639	1639
		CatBoost-UQ	0.360 ± 0.018	0.999 ± 0.001	0.457 ± 0.012	**0.0072 ± 0.0003**	**0.0093 ± 0.0004**	1639	1639
		E-NGBoost (NIG)	0.861 ± 0.047	0.794 ± 0.251	0.903 ± 0.024	0.0170 ± 0.0045	0.0235 ± 0.0060	1639	1639

Future work.

1. **More principled objectives.** Develop training objectives (or constraints) that avoid incentivizing “inflate uncertainty to reduce loss”, ideally with a clear testable criterion and a theoretically grounded selection rule.
2. **Principled density priors and calibration.** Connect $\nu_{\text{target}}(x)$ to an explicit prior over evidence, or derive inference-time calibration rules with guarantees under shift assumptions.
3. **Broader baselines and diagnostics.** Adding stronger tabular-UQ baselines (additional conformal variants, alternative distributional predictors) and richer calibration diagnostics would further strengthen the empirical story.

7 CONCLUSION

We presented E-NGBoost, a practical integration of NGBoost with NIG evidential regression for tabular uncertainty quantification. The resulting Student-t predictive marginal provides an analytic uncertainty decomposition into aleatoric and epistemic components. To improve robustness and to encourage lower evidence in low-density regions, we incorporated density-aware evidence regularization using an Isolation Forest density proxy fit on the training split only.

Using a conservative, leakage-controlled evaluation protocol with proper scoring rules and calibration diagnostics, E-NGBoost is competitive with NGBoost in-distribution and can provide useful uncertainty ranking signals for feature-aligned OOD detection and selective prediction in some settings. Clear limitations are also observed, including hyperparameter sensitivity and shift-dependent performance, which motivates future work on more principled objectives and calibration strategies.

REPRODUCIBILITY

All experiments are run from the repository in `Workshop_Evidential_Boosting/`. Commands below match the ones used to generate the tables in this draft.

ID metrics + calibration diagnostics.

```
python3 -m Workshop_Evidential_Boosting.src.run_mvp_loop \
```

```
  --datasets concrete_compressive_strength energy_efficiency boston kin8nm diabetes \
  --n-splits 5 --test-size 0.2 --n-estimators 300 --ensemble-size 5 \
  --crps-samples 200 --include-catboost --save-splits --run-tag paper_v1
```

Interval baselines (SplitConformal/CQR/QRF) with an explicit calibration split.

```
python3 -m Workshop_Evidential_Boosting.src.run_mvp_loop \
  --datasets concrete_compressive_strength energy_efficiency boston kin8nm diabetes \
  --n-splits 5 --test-size 0.2 --cal-size 0.2 --include-catboost \
  --include-conformal --include-cqr --include-qrf --save-splits \
  --run-tag interval_cal20_v1
```

Energy calibration-CRPS tuning.

```
python3 -m Workshop_Evidential_Boosting.src.tune_energy_lambdas \
  --dataset energy_efficiency --n-splits 5 --test-size 0.2 --cal-size 0.2 \
  --n-estimators 300 --crps-samples 200 --run-tag energy_tuned_crps_v1
```

OOD table runs (example).

```
python3 -m Workshop_Evidential_Boosting.src.run_ood_loop \
  --dataset boston --shift tail --n-splits 5 --lambda-d 0.1 \
  --nu-target-scale 10 --enig-infer-if --run-tag boston_tail_dens10_if_v1
```

A Core Mathematical Formulation

A.1 NIG hierarchy and Student-t marginal

E-NGBoost predicts $(\gamma, \nu, \alpha, \beta)$ per input x and uses the standard NIG hierarchy:

$$\sigma^2 \sim \mathrm{InvGamma}(\alpha, \beta), \qquad \mu \mid \sigma^2 \sim \mathcal{N}(\gamma, \sigma^2/\nu), \qquad y \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2).$$

Marginalizing (μ, σ^2) yields a Student-t predictive distribution:

$$y \mid x \sim \text{Student-t}_{2\alpha}\Big(\gamma,\ s^2 = \frac{\beta(1+\nu)}{\nu\alpha}\Big).$$

We use the common analytic variance decomposition (diagnostic interpretation only):

$$\mathrm{Var}_{al} = \mathbb{E}[\sigma^2] = \frac{\beta}{\alpha - 1}, \qquad \mathrm{Var}_{ep} = \mathrm{Var}[\mu] = \frac{\beta}{\nu(\alpha-1)}, \qquad \mathrm{Var}(y) = \mathrm{Var}_{al} + \mathrm{Var}_{ep}.$$

A.2 Closed-form Student-t NLL used in training

Let $\delta = y - \gamma$, $\Omega = 2\beta(1+\nu)$, and $T = \nu\delta^2 + \Omega$. The Student-t marginal NLL used in our implementation can be written as:

$$\mathcal{L}^{\mathrm{NLL}} = \tfrac{1}{2}\log(\pi/\nu) \ - \ \alpha \log \Omega \ + \ (\alpha + \tfrac{1}{2}) \log T \ + \ \log \Gamma(\alpha) \ - \ \log \Gamma(\alpha + \tfrac{1}{2}).$$

A.3 Natural gradient and diagonal Fisher (log-parameter space)

NGBoost updates distribution parameters with a natural gradient preconditioner. We work in log-parameter space for constrained parameters:

$$\phi = (\gamma, \log \nu, \log(\alpha - 1), \log \beta),$$

using the chain rule $\partial/\partial \log \nu = \nu\, \partial/\partial\nu$, $\partial/\partial \log(\alpha-1) = (\alpha-1)\, \partial/\partial\alpha$, and $\partial/\partial \log \beta = \beta\, \partial/\partial\beta$.

We use a diagonal approximation $I(\phi) \approx \mathrm{diag}(I_1, \dots, I_4)$ and compute the natural gradient as $\tilde{g} = I(\phi)^{-1}\nabla_\phi \mathcal{L}$. The resulting closed-form diagonal entries are:

$$I_{\gamma\gamma} = \frac{\nu(2\alpha+1)}{2\beta(1+\nu)}, \qquad I_{\log\nu,\log\nu} = \frac{2\alpha+1}{2(1+\nu)}, \qquad I_{\log(\alpha-1),\log(\alpha-1)} = (\alpha-1)^2\big(\psi'(\alpha) - \psi'(\alpha + \tfrac{1}{2})\big), \qquad I_{\log\beta,\log\beta} = \alpha,$$

where $\psi'(\cdot)$ is the trigamma function.

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